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Does Imputation Change the Winner? Missing Value Treatment in Alternative Voting Systems Using the France 2022 Electoral Experiment

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Abstract

This thesis investigates how different treatments of missing values affect electoral outcomes in alternative voting systems using data from the Voter Autrement 2022 French presidential election experiment. Incomplete ballots arise in Range Voting whenever participants choose not to evaluate all candidates, yet the methodological consequences of handling these missing values have received comparatively little systematic attention. The original experiment addressed this issue by assigning the lowest possible score to every unevaluated candidate. This thesis examines whether that choice is robust to alternative missing-data treatments.

Six imputation strategies are applied to the four Range Voting datasets, ranging from simple approaches such as lowest-rank imputation and complete-case deletion

to statistically more sophisticated methods including multiple imputation. Approval Voting and Borda Count serve as stable, missing-value-free baselines. The analysis evaluates whether different imputation strategies alter candidate rankings or the identity of the winner.

Multiple imputation is implemented using predictive mean matching with five completed datasets. Candidate mean scores are pooled using Rubin's rules, while rankings are evaluated separately within each completed dataset to assess the stability of electoral outcomes under imputation uncertainty. Additional robustness analyses examine the sensitivity of cluster-based and k-nearest neighbour imputation to alternative values of the tuning parameter k.

The analysis finds that the overall electoral outcome remains highly robust across the imputation strategies considered. While minor changes occur among a small number of closely ranked candidates, the overall winner remains stable throughout the analyses. The findings suggest that, for this election, the robustness of the winner is primarily attributable to the substantial margin separating the leading candidate from the remainder of the field rather than to the irrelevance of missing-data treatment itself. The thesis therefore concludes that transparent and methodologically justified handling of missing values remains an essential component of empirical electoral research, particularly in more closely contested elections where imputation choices may have greater practical consequences.

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1 Introduction

The analysis of alternative voting systems has become increasingly significant in political science and social choice theory. France, in particular, has emerged as a key case for experimental electoral research through the *Voter Autrement* project, which investigates citizen behaviour under various voting rules beyond traditional plurality voting. Conducted alongside several French presidential elections since 2002, these experiments have generated unique datasets containing voter evaluations across multiple electoral systems (Delemazure & Bouveret, 2024).

This bachelor's thesis focuses on the 2022 French presidential election dataset and examines how different treatments of missing values in Range Voting affect electoral outcomes. Range Voting explicitly allows voters to abstain from evaluating individual candidates, producing incomplete ballots, a problem the original experiment handled by assigning the lowest possible score to all unevaluated candidates (Delemazure & Bouveret, 2024). Whether this choice distorts the final rankings relative to more sophisticated alternatives is the central question this thesis investigates.

The treatment of missing values poses a significant methodological challenge (Little & Rubin, 2002). Different assumptions about NAs can substantially affect candidate rankings, aggregate outcomes, and the interpretation of voter preferences. Assigning missing values to the lowest possible rank assumes that unevaluated candidates are least preferred, an assumption that conflates unfamiliarity with disapproval. Complete-case deletion of incomplete ballots reduces the sample size and may introduce selection bias. Alternative approaches, such as mean imputation, cluster-based imputation, or k-nearest-neighbour imputation, rely on statistical assumptions about voter similarity and preference structures, each encoding a different theory of what it means to leave a candidate unevaluated.

This challenge is particularly relevant because prior research on voting systems has concentrated primarily on the theoretical properties of electoral rules, leaving the practical implications of missing data treatment comparatively underexplored (Baujard et al., 2014; Baujard, Igersheim & Lebon, 2024). At the same time, recent literature increasingly emphasises the importance of empirical preference structures and voter heterogeneity in understanding electoral outcomes.

The discussion paper Dichotomous Preferences: Concepts, Measurement, and Evidence by Barbaro and Kurella (2025) illustrates the growing role of experimental electoral datasets and cluster-analytical approaches in understanding voter preferences. That paper focuses primarily on the Continuous Opinion Form included in the France22 experiment, in which participants evaluated candidates using a slider on a scale from 0 to 100. Because the slider's default position corresponded to the neutral value of 50, a recorded value of 50 may represent either a genuine neutral opinion or a non-response, making it impossible to distinguish actual neutral evaluations from missing data. This highlights a broader methodological ambiguity that arises when missing information is not explicitly identifiable in electoral datasets, and underscores the value of working with systems, such as Range Voting, where abstention is unambiguously recorded.

This thesis addresses that gap by working with the Range Voting datasets, which contain explicit, unambiguous missing values and a formal aggregation rule that makes it possible to trace how different imputation choices affect the final ranking.

The relevance of this question operates on two levels, though both should be read with the scope of a single case study in mind. Methodologically, if the results of an electoral experiment depend on a preprocessing decision that is rarely made explicit and almost never compared systematically, the apparent objectivity of quantitative electoral analysis can be weaker than it looks, at least in elections where imputation choices turn out to be consequential, which, as Section 5 and Section 6.3 discuss, this particular election's large margin of victory does not allow it to demonstrate directly. Making this choice transparent and subjecting it to systematic comparison across strategies is nonetheless a contribution to the reproducibility of experimental electoral research, independent of whether the winner happens to change in any one dataset. Democratically, these experiments are increasingly used to inform normative arguments about which voting systems best represent voter preferences; where such arguments rest on findings that would shift under a different, equally defensible missing-data treatment, their evidential basis is weaker than it appears. This thesis treats missing data handling as a substantive methodological decision rather than a technical afterthought, while also being explicit that the France22 sample is a self-selected, unauthenticated online sample whose aggregate outcome diverges substantially from the actual 2022 election result (Section 2.1). This limits how far any normative claim about democratic representation can be drawn from it, a point revisited in Section 6.3.

The central objective of this thesis is therefore to examine how sensitive electoral outcomes in Range Voting are to different treatments of missing values, and specifically whether the lowest-rank imputation applied in the original experiment produces results that are robust to methodological alternatives. Six imputation strategies are applied and compared against Approval Voting and Borda Count as stable, missing-value-free baselines, an experimental design that allows for a structured assessment of whether missing data treatment systematically alters electoral outcomes.

Methodologically, the thesis integrates political science, social choice theory, and quantitative data analysis, implemented in R using the Positron environment. The analysis draws on the France22 dataset (Delemazure & Bouveret, 2024), which includes responses from 2,308 participants who evaluated 12 candidates across multiple voting systems during the 2022 French presidential election.

To guide the empirical analysis, four expectations are formulated and tested in this thesis. These expectations are introduced in full in Section 4.2 and revisited against the empirical findings in Section 6.1.

The remainder of the thesis is structured as follows. Section 2 introduces the *Voter Autrement* experiment and the voting systems included in the dataset. Section 3 discusses the voting systems used as baselines for comparison. Section 4 presents the theoretical and methodological background on missing data and the six imputation strategies, together with the expectations guiding the analysis. Section 5 presents the empirical results, including the comparison with the Approval Voting and Borda Count baselines. Section 6 discusses the findings in light of the expectations formulated beforehand, before Section 7 concludes.

2 The Voter Autrement 2022 Experiment

2.1 Background and Design

The Voter Autrement project is a long-running series of experimental voting studies conducted alongside French presidential elections. Since its beginning in 2002, the project has spanned five consecutive elections (2002, 2007, 2012, 2017, and 2022) across which a total of 12 separate experiments were carried out. The 2022 iteration, which forms the empirical basis of this thesis, is thus embedded in a broader longitudinal research programme systematically testing alternative electoral methods over time (Baujard et al., 2014; Baujard, Igersheim & Lebon, 2024).

The 2022 experiment was conducted in two parallel formats: an in situ experiment at polling stations in Strasbourg, and an online experiment accessible to the general public via the platform *vote.imag.fr*. The online component, documented in detail by Delemazure and Bouveret (2024), ran from April 8th to May 7th, 2022, spanning the period between the first and second rounds of the presidential election. It attracted 2,308 participants who completed the full experimental protocol.

Participation was voluntary and open to anyone with internet access. The experiment was disseminated through academic mailing lists and social media. To prevent automated participation, a CAPTCHA was included at the entry point. Participants were not required to authenticate themselves, meaning multiple participation cannot be entirely ruled out, a limitation the authors acknowledge explicitly (Delemazure & Bouveret, 2024). Despite this, the dataset is widely regarded as a high-quality source for experimental electoral research, and it's open availability under the Open Database License makes it particularly well-suited for replication and methodological comparison.

2.2 The Dataset

The dataset resulting from the online experiment is publicly available on Zenodo (doi:10.5281/zenodo.10 and consists of 15 CSV files. Each row in the data files corresponds to one participant, identified by a consistent numerical ID across files. The main components relevant to this thesis are the individual voting rule files and the voters.csv file, which contains demographic information including age group, gender, level of education and socio-professional category.

The Range Voting files, which constitute the primary focus of this thesis, are available in four variants corresponding to different score scales: (0, 1, 2), (0, 1, 2, 3), (−1, 0, 1), and (−1, 0, 1, 2). Each file contains scores assigned by participants to all 12 candidates. Where a participant chose not to evaluate a candidate, the corresponding cell is empty. In the original study, these empty cells were treated as lowest-rank entries. That is, the minimum value of the respective scale was assigned to all missing observations (Delemazure & Bouveret, 2024). This thesis takes that decision as its starting point and systematically examines whether the electoral outcomes it produces are robust to alternative imputation choices.

The apparent asymmetry in participant counts across the four Range Voting files is fully explained by the experiment's randomisation design, recorded in `voters.csv` via four presentation slots (`rule_1` through `rule_4`) assigned to each participant. Slot 1 draws from Approval Voting, `scores_3_neg`, and `scores_4_neg`, while Slot 3 draws from all four Range Voting scale variants. Participants assigned `scores_3_neg` or `scores_4_neg` in Slot 1 are always paired with the corresponding positive-scale variant (`scores_3_pos` or `scores_4_pos`, respectively) in Slot 3, whereas participants assigned Approval Voting in Slot 1 are paired with one of `scores_3_neg`, `scores_4_neg`, or `scores_4_pos` in Slot 3 (never `scores_3_pos`). This produces three mutually exclusive experimental arms of 1,382 (Approval-first), 475 (three-point-first), and 451 (four-point-first) participants, which sum exactly to the reported total of 2,308. This design explains, for instance, why `scores_3_pos` shares almost all of its participants with `scores_3_neg` but none with Approval Voting, and why `scores_3_neg` (903 participants) is essentially the union of its 475 three-point-arm and 428 Approval-arm respondents.

3 Voting Systems and Baseline Selection

The France22 experiment included six distinct voting systems. Not all of them are equally suited for the imputation analysis at the heart of this thesis, and it is therefore important to explain which systems are used and why others are excluded. The selection is not arbitrary but follows directly from the methodological requirements of the research design: the analysis requires systems that produce numerical scores per candidate, apply a formal aggregation rule, and, in the case of the primary focus, generate identifiable missing values that can be subjected to different imputation strategies. On this basis, three systems are included: Range Voting as the primary focus and Approval Voting and Borda Count as stable, missing-value-free baselines.

3.1 Range Voting

Range Voting is a scoring-based electoral method in which each voter assigns a numerical score to every candidate from a predefined scale. The candidate with the highest aggregate score wins. In the France22 experiment, Range Voting was included in four variants using different scales: (0, 1, 2), (0, 1, 2, 3), (-1, 0, 1), and (-1, 0, 1, 2). Participants were not required to evaluate every candidate; if they chose to abstain on a particular candidate, no score was recorded, resulting in a missing value.

This explicit allowance for abstention is what makes Range Voting the methodologically appropriate focus for this thesis. Missing values in these datasets are unambiguous. They represent a deliberate choice not to score, clearly distinguishable from any particular score value. The six imputation strategies applied in Section 4 each make different assumptions about what this non-response means, and their results are compared to determine whether the original lowest-rank choice was consequential.

3.2 Approval Voting

Approval Voting is the simplest of the systems included in the experiment. Each voter either approves (1) or disapproves (0) of each candidate; every candidate must receive a response. The winning candidate is the one approved by the largest number of voters. Because the binary structure of the ballot leaves no room for abstention, the Approval Voting dataset contains no missing values. This makes it a methodologically stable and transparent baseline against which Range Voting outcomes can be compared (Baujard et al., 2014; Baujard et al., 2018).

3.3 Borda Count

In the Borda Count, as implemented in the France22 experiment, participants were asked to rank exactly four candidates. The experimental interface prevented progression to the next step unless precisely four candidates had been ranked, ensuring complete data. The top-ranked candidate received four points, the second three, the third two, and the fourth one; all remaining candidates received zero points. Like Approval Voting, the Borda Count dataset contains no missing values, making it a second reliable baseline for comparison.

3.4 Remaining Systems

The remaining systems included in the experiment: Instant Runoff Voting, Majority Judgement, the Continuous Opinion Form, and Pairwise Comparisons are not included in the analysis.

Instant Runoff Voting is an elimination-based system that produces no aggregate numerical score per candidate, making direct comparison with imputed Range Voting outcomes methodologically problematic.

Majority Judgement uses ordinal labels rather than numerical scores, which are incompatible with the score-based imputation strategies applied here.

The Continuous Opinion Form, while producing numerical values, conflates genuine neutral responses with non-responses due to the default slider position of 50 (Barbaro & Kurella, 2025). Pairwise Comparisons produce only binary preference relations between candidate pairs and therefore yield no comparable overall ranking.

4 Missing Value Imputation Strategies

4.1 The Problem of Missing Data in Range Voting

Range Voting is unique among the systems included in the France22 experiment in that it explicitly permits abstention. Voters may choose not to evaluate individual candidates, producing observable gaps in the dataset. Each of the four Range Voting variants contains missing values wherever a participant did not assign a score to a given candidate. In the dataset for the (-1, 0, 1) scale, for instance, a total of 702 missing values are recorded across 898 participants evaluating 12 candidates, representing a non-trivial proportion of the data.

The methodological challenge this poses is well established in the statistical literature (Little & Rubin, 2002): the appropriate treatment of missing values depends fundamentally on why they are missing. In the context of experimental voting, a voter who does not evaluate a candidate may be expressing genuine indifference, or may simply be unfamiliar with the candidate in question. Confusing these two interpretations, as lowest-rank imputation implicitly does, risks systematically distorting the final aggregate scores of less well-known candidates. The France22 experiment addressed this problem by applying lowest-rank imputation as its default strategy, assigning the minimum scale value to all missing entries (Delemazure & Bouveret, 2024). While this choice is transparent and easy to implement, it encodes a strong and potentially contestable assumption about voter intent. Whether it produces results that are robust to methodologically alternative approaches is the central question this thesis investigates.

Before turning to the individual strategies, it is useful to introduce three concepts from the statistical literature that classify missing data according to why it is missing, since several of the strategies discussed below rest on different assumptions about this underlying mechanism (Little Rubin, 2002, van Buuren 2018).

Missing Completely at Random (MCAR): the probability that a value is missing is entirely unrelated to any observed or unobserved variable. In the electoral context, this would mean that whether a voter evaluates a candidate is pure chance, unrelated to the voter's characteristics or the candidate's profile.

Missing at Random (MAR): the probability that a value is missing may depend on other observed variables, but not on the missing value itself. For example, whether a voter evaluates a given candidate may depend on how that voter scored other candidates, therefore information that is observed, even though the score that would have been given is not.

Missing Not at Random (MNAR): the probability that a value is missing depends on the missing value itself. For instance, if voters were systematically less likely to evaluate candidates they disliked. MNAR is the most difficult mechanism to address statistically, since the very information needed to correct for it is unobserved.

These three mechanisms matter because different imputation strategies are only valid, in a statistical sense, under different assumptions about which mechanism generated the missing data. Complete-case deletion (Section 4.4) is unbiased only under MCAR; multiple imputation (Section 4.8) is valid under the weaker and more realistic MAR assumption. None of the six strategies considered in this thesis fully addresses MNAR, which would require explicit modelling of the relationship between a candidate's true (unobserved) score and the probability that it was left unevaluated, a task beyond the scope of the present analysis.

Beyond lowest-rank imputation, five further strategies are considered in this thesis. Each encodes a different assumption about the nature of the missing data and makes different trade-offs between statistical stiffness, sample retention and sensitivity to distributional assumptions. Taken together, the six strategies span a spectrum from the simplest possible approach to the most statistically sophisticated, allowing for a systematic comparison of their consequences for electoral outcomes.

4.2 Expectations

Drawing on the methodological literature on missing data and on prior Voter Autrement research, four expectations guide the analysis presented in Section 5.

E1 – Frontrunner stability: The leading candidate is expected to be robust across all six imputation strategies, consistent with prior findings that plurality frontrunners tend to win across a wide range of electoral rules and missing data treatments (Baujard et al., 2014; Baujard, Igersheim & Lebon, 2024).

E2 – Lower-rank sensitivity: Mid-field and lower-ranked candidates are expected to be considerably more sensitive to the choice of imputation strategy than the frontrunner, since their aggregate scores are likely to lie closer together and therefore more easily reordered by small changes in imputed values.

E3 – Lowest-rank bias: Lowest-rank imputation is expected to systematically disadvantage candidates who are frequently left unevaluated due to unfamiliarity rather than genuine disapproval, since the strategy assigns such candidates the minimum possible score regardless of the reason for non-response (Baujard et al., 2018).

E4 – MI as most defensible: Multiple imputation is expected to produce the most statistically defensible results among the six strategies, as it is the only method that explicitly accounts for the uncertainty inherent in missing data rather than substituting a single point estimate (Rubin, 1987; van Buuren, 2018).

These four expectations are revisited in light of the empirical results in Section 5 and discussed in detail in Section 6.

4.3 Lowest-Rank Imputation

Lowest-rank imputation is the simplest of the six strategies and the one applied in the original Voter Autrement 2022 experiment (Delemazure & Bouveret, 2024). The underlying assumption is that a voter who did not evaluate a candidate implicitly regards that candidate as least preferred. All missing values are therefore replaced with the minimum value of the respective scale: 0 for the (0, 1, 2) and (0, 1, 2, 3) variants, and -1 for the (-1, 0, 1) and (-1, 0, 1, 2) variants. This substitution rule reproduces the convention adopted by the original experiment (Delemazure & Bouveret, 2024) rather than a procedure recommended in the general missing-data literature; Little and Rubin (2002) is cited elsewhere in this thesis for the broader statistical framework of missing-data mechanisms introduced in Section 4.1, not as an endorsement of worst-value substitution specifically.

While this approach is transparent and easy to implement, its assumption is strong. Research on voter knowledge suggests that candidates are frequently left unevaluated not because they are disliked, but because they are unfamiliar (Baujard et al., 2018). Assigning such candidates the minimum score conflates unfamiliarity with disapproval, which may systematically disadvantage lesser-known candidates regardless of their actual support among informed voters.

The function identifies the minimum value of the score matrix and replaces all NA entries with that value. Because the minimum is computed across the entire dataset rather than per column, it correctly handles all four scale variants without requiring manual adjustment.

4.4 Complete-Case Deletion

Complete-case deletion removes all ballots that contain at least one missing value, retaining only those participants who evaluated every candidate. Rather than filling gaps with estimated values, this strategy simply restricts the analysis to fully observed data. It is therefore the only approach among the six that does not impute anything (Little & Rubin, 2002).

The main advantage of complete-case deletion is its conceptual simplicity: the analysis is based entirely on observed preferences, with no assumptions about what missing values might have been. However, this comes at a cost. First, the effective sample size is reduced, which can affect the stability of aggregate scores for candidates with many missing entries. Second, and more importantly, complete-case deletion is unbiased only under the MCAR mechanism introduced in Section 4.1 (Little Rubin, 2002). In the context of electoral experiments, this assumption is likely violated: voters who are less politically informed, or who are unfamiliar with outsider candidates, are systematically more likely to produce incomplete ballots, meaning the missingness is more plausibly MAR than MCAR. The resulting dataset retains only those rows with no missing values across all 12 candidate columns. For the $(-1, 0, 1)$ scale dataset, this results in a noticeably smaller sample, and the degree of reduction itself provides useful descriptive information about the extent to which incomplete responses characterize the data.

4.5 Mean Imputation

Mean imputation replaces each missing value with the arithmetic mean of the observed scores for that candidate across all complete responses. The underlying assumption is that the average score reflects the typical preference of the electorate towards a given candidate, and that a voter who left a candidate unevaluated would, on average, have scored that candidate similarly to other voters (Little & Rubin, 2002).

Mean imputation preserves the overall mean of each candidate's score distribution and is straightforward to implement. However, it is well known to have significant statistical drawbacks: by replacing missing values with a constant, it artificially reduces the variance of the imputed variable and distorts the correlation structure between candidates. In the electoral context, this means that mean imputation tends to produce moderate, centrist outcomes, by pulling all candidates slightly towards the average, which may mask genuine voter heterogeneity and polarization (Little & Rubin, 2002). The function computes the column mean from all non-missing values and substitutes it for each NA in that column. Because the mean is computed separately per candidate, the function correctly handles the different popularity levels of individual candidates rather than applying a single global value.

4.6 Cluster-Based Imputation

Cluster-based imputation extends the logic of mean imputation by computing imputed values not from the full sample but from demographically similar subgroups. The assumption is that voters with similar characteristics, in terms of age, gender, education, and socio-professional category, tend to exhibit similar voting behavior, and that missing values are better estimated from within-group means than from global averages (Little & Rubin, 2002). This approach is inspired by the cluster-analytical framework developed by Barbaro and Kurella (2025). While their study uses k-means clustering to analyze voter preference structures, the present thesis adapts the same underlying idea for missing value imputation by assigning incomplete observations to the nearest cluster and using the corresponding cluster center to estimate missing values.

The implementation merges each Range Voting file with the demographic variables recorded in voters.csv (age group, gender, level of education, and socio-professional category), treating unanswered demographic questions ("nspp", "prefer not to say") or genuinely missing entries as their own category rather than discarding those respondents. Because these variables are categorical rather than continuous, clusters are formed using Gower's distance, which accommodates mixed and categorical data, combined with k-medoids clustering rather than k-means, which assumes continuous, Euclidean-scaled input and is not appropriate for categorical demographic variables. For each incomplete ballot, the missing scores are replaced with the mean of the observed scores for that candidate within the respondent's demographic cluster; if no observed scores for a given candidate exist within a cluster, the global column mean is used as a fallback.

The value of k is set to five, for comparability with the k-nearest-neighbour strategy in Section 4.7. In practice, the four categorical variables available (age, gender, education, socio-professional category) produce a limited number of common combinations, so the resulting clusters are highly imbalanced: one dominant cluster typically contains the large majority of respondents, alongside several much smaller clusters representing less common demographic combinations. This limits how much genuinely subgroup-specific signal the strategy can capture in practice, a point revisited in Section 6.3.

4.7 k-Nearest Neighbor Imputation

k-nearest neighbor (kNN) imputation estimates each missing value by identifying the k voters most similar to the incomplete voter in terms of their observed scores and averaging their values for the missing candidate. Unlike cluster-based imputation, which assigns all members of a demographic group the same imputed value, kNN imputation is voter-specific: it finds the nearest neighbors in the actual score space rather than in a pre-defined cluster structure. The method was formalized by Troyanskaya et al. (2001) and has since been widely adopted in contexts where missing data patterns are likely to reflect systematic but individual-level variation.

The key assumption of kNN imputation is that voters with similar preferences for the candidates they did evaluate will also have similar preferences for the candidates they did not evaluate. This assumption is plausible in the electoral context, where voting behavior tends to be structured by underlying ideological dimensions that predict preferences across candidates simultaneously. kNN imputation is therefore expected to preserve more of the correlation structure of the data than mean or cluster-based imputation.

The function iterates over each candidate column with missing values, trains a kNN model on the complete observations for that column, and predicts the missing scores using the remaining candidates as predictors. The parameter k is set to five, meaning that the missing value is estimated based on the five most similar voters. All five neighbors contribute equally to the prediction. As with cluster-based imputation, k is set to 5 by default.

4.8 Multiple Imputation

Multiple imputation (MI) is the most statistically sophisticated of the six strategies considered in this thesis. Rather than replacing each missing value with a single estimate, MI generates several plausible completed versions of the dataset by repeatedly drawing values from the posterior predictive distribution implied by the observed data. Each completed dataset is analysed independently, allowing uncertainty introduced by missing data to be propagated through the analysis rather than ignored (Rubin, 1987).

The validity of multiple imputation relies on the Missing at Random (MAR) assumption introduced in Section 4.1.Under MAR, the probability that a value is missing may depend on observed information but not on the missing value itself. In the context of the France22 experiment, this assumption is more plausible than MCAR because a participant's decision not to evaluate a candidate is likely to be associated with observed characteristics of the ballot, such as evaluations of other candidates, rather than being entirely random.

Multiple imputation is implemented using the mice package (van Buuren, 2018) with predictive mean matching (PMM). PMM is particularly suitable for bounded voting scales because every imputed value is sampled from an observed value rather than generated through unrestricted regression prediction. Consequently, the completed datasets preserve the discrete structure and observed range of the original Range Voting scores. Five completed datasets ($m = 5$) are generated for each of the four Range Voting datasets. Candidate mean scores are calculated separately within every completed dataset together with their associated sampling variances. Rubin's pooling rules (Rubin, 1987) are then applied to combine these candidate-level estimates by incorporating both the average within-imputation variance and the between-imputation variance. This produces pooled estimates of candidate mean scores together with pooled standard errors and confidence intervals that explicitly reflect uncertainty due to missing data.

Because electoral rankings are ordinal outcomes rather than continuous statistical parameters, the rankings themselves are not directly pooled. Instead, candidate rankings are calculated independently within each completed dataset, after which the pooled candidate mean scores are used to construct an overall MI ranking. In addition, the rankings and winners from all five completed datasets are retained and compared directly. This makes it possible to evaluate not only the pooled ranking but also whether the winner and the full ordering of candidates remain stable across individual imputations. The random seed is fixed at 42 to ensure complete reproducibility of the imputation process.

5 Analysis and Results

This section presents the results of applying all six imputation strategies to the four Range Voting datasets. The analysis proceeds in three steps: first, the candidate rankings produced by each strategy are compared within each dataset; second, the stability of the overall winner is assessed across strategies and scales; and third, the sensitivity of individual candidates' rankings to the choice of imputation method is examined in detail.

5.1 Candidate Rankings by Strategy

Across all four Range Voting datasets and all six imputation strategies, the top three candidates are identical: Jean-Luc Mélenchon ranks first, Yannick Jadot second, and Philippe Poutou third, regardless of which scale or which imputation method is applied.

The lower end of the ranking is similarly stable: Eric Zemmour ranks last (12th) and Marine Le Pen second-to-last (11th) across every strategy and every scale variant. The total scores assigned to each candidate do differ meaningfully between strategies. For instance, Mélenchon's aggregate score in the (0, 1, 2) dataset (scores_3_neg) ranges from 576 under complete-case deletion to 732 under mean imputation, but these differences in magnitude do not translate into differences in rank.

5.2 Winner Stability

Across all 24 combinations of dataset and imputation strategy (four scales $\times$ six strategies), Jean-Luc Mélenchon is identified as the winner in every single case. No imputation strategy, on any scale, produces a different overall winner. This finding directly addresses the question motivating this thesis: the lowest-rank imputation applied in the original *Voter Autrement* experiment (Delemazure & Bouveret, 2024) does not appear to have distorted the identity of the winning candidate relative to more statistically sophisticated alternatives such as multiple imputation.

This result should not be read as evidence that imputation strategy is irrelevant. Rather, it suggests that the margin of victory in this particular dataset was large enough that no plausible treatment of missing values was capable of overturning it. Whether this robustness generalizes to closer elections, or to electoral experiments with a higher proportion of missing data, remains an open question that the discussion in Section 6 returns to.

5.3 Multiple Imputation Robustness

For each of the four Range Voting datasets, five completed datasets were generated using predictive mean matching. Candidate rankings were calculated separately within each completed dataset before candidate mean scores were pooled using Rubin's rules.

The results demonstrate complete winner stability across all imputations. Table 1 summarizes the winner obtained in each completed multiple-imputation analysis.

Dataset	Winner in all 5 imputations
scores_3_neg	Jean-Luc Mélenchon (5/5)
scores_3_pos	Jean-Luc Mélenchon (5/5)
scores_4_neg	Jean-Luc Mélenchon (5/5)
scores_4_pos	Jean-Luc Mélenchon (5/5)

Table 1: *Winner across all completed multiple-imputation datasets*

Thus, the stability of the winner is not merely a consequence of pooling but is observed consistently in every individual completed dataset.

Candidate rankings are similarly robust. For the overwhelming majority of candidates, the rank remains identical across all five imputations. Where variation occurs, it is limited to a single adjacent rank position, indicating only minimal between-imputation uncertainty. No candidate exhibits substantial instability across imputations.

These findings strengthen the interpretation of the pooled MI results. Rather than relying solely on pooled point estimates, the analysis demonstrates directly that repeated plausible completions of the missing data produce virtually identical electoral outcomes. For the present dataset, uncertainty introduced by the imputation process is therefore small relative to the observed differences between candidates.

5.4 Rank Sensitivity

While the winner is stable across all strategies, a small number of candidates in the middle of the ranking do exhibit minor rank shifts. Out of 48 candidate-dataset combinations (12 candidates $\times$ 4 datasets), only four show any sensitivity at all to the choice of imputation strategy, and in each case the shift is limited to a single rank position. Table 2 summarizes these cases.

Dataset	Candidate	Min Rank	Max Rank
scores_4_neg	Emmanuel Macron	7	8
scores_4_neg	Jean Lassalle	7	8
scores_4_pos	Emmanuel Macron	6	7
scores_4_pos	Nathalie Arthaud	6	7

Table 2: *Candidates exhibiting rank sensitivity across imputation strategies*

In both datasets, only two neighbouring candidates effectively swap positions depending on the imputation method used: Macron and Lassalle in the $(-1, 0, 1, 2)$ scale, and Macron and Arthaud in the $(0, 1, 2, 3)$ scale. Notably, all other candidates, including the top three and bottom three, retain identical ranks across every strategy. This pattern suggests that rank sensitivity to imputation choice, where it exists in this dataset, is concentrated among candidates whose aggregate scores are closely clustered.

Additional robustness analyses were conducted for the two imputation methods that require a user-defined tuning parameter. Cluster-based imputation was repeated for $k = 3, 5, 7$, and 9, while k-nearest neighbour imputation was repeated for $k = 3, 5, 7, 9$, and 11. Across all tested values of k, Jean-Luc Mélenchon remained the winner and no candidate changed rank in any of the four Range Voting datasets. Table 3 summarizes these results.

Method	Tested k values	Winner changed	Rank changes
Cluster-based imputation	3, 5, 7, 9	No	None
kNN imputation	3, 5, 7, 9, 11	No	None

Table 3: *Sensitivity of rankings to alternative values of k*

The observed rank differences therefore reflect differences between imputation strategies rather than arbitrary parameter choices within individual methods.

5.5 Comparison with Approval Voting and Borda Count

Because Approval Voting and Borda Count contain no missing values, they serve as stable reference points against which the imputed Range Voting rankings can be evaluated. Both baseline systems independently confirm Jean-Luc Mélenchon as the winner, with a substantial margin over the second-placed candidate: 984 points to Jadot's 812 under Approval Voting, and 2,115 to 1,613 under Borda Count. This corroborates the winner stability finding reported in Section 5.2 using an entirely independent methodological route, one that requires no imputation at all.

Table 2 compares the full ranking produced by Approval Voting and Borda Count against the Range Voting ranking under lowest-rank imputation, the strategy applied in the original experiment.

The agreement between the three systems is striking at both extremes of the ranking: the top three candidates and bottom two candidates are identical across all three methods. The middle of the ranking, however, reveals small but consistent discrepancies. Roussel and Hidalgo swap places between Approval Voting and Borda Count, while Macron and Arthaud occupy different relative positions in Range Voting than in either baseline. Most notably, Nicolas Dupont-Aignan ranks 10th under Range Voting but last (12th) under both Approval Voting and Borda Count. The substantial divergence holds across all four Range Voting scales and all six imputation strategies, suggesting that it reflects a genuine difference in how Range Voting aggregates preferences for this candidate rather than an artefact of missing data treatment.

This divergence is informative for interpreting the central robustness finding of this thesis. The fact that Dupont-Aignan's rank differs systematically between Range Voting and the two baselines, regardless of imputation strategy, indicates that this particular discrepancy is a property of the voting rule itself, rooted in how Range Voting allows voters to express graded preferences rather than binary approval or fixed-position rankings, and not a consequence of how missing values were treated. This reinforces the

Candidate	Approval Voting	Borda Count	Range Voting
Melenchon	1	1	1
Jadot	2	2	2
Poutou	3	3	3
Roussel	4	5	4
Hidalgo	5	4	5
Macron	6	6	7
Arthaud	7	7	6
Lassalle	8	8	8
Pecresse	9	9	9
Le Pen	10	10	11
Zemmour	11	11	12
Dupont-Aignan	12	12	10

Table 4: *Candidate ranking comparison: Approval Voting, Borda Count, and Range Voting*

broader conclusion that imputation strategy affects outcomes only at the margin, while the choice of voting system itself remains the dominant source of variation in candidate rankings.

6 Discussion

6.1 Revisiting the Expectations

The four expectations formulated in Section 4.2 can now be assessed against the empirical results presented in Section 5.

E1 – Frontrunner stability (confirmed): Jean-Luc Mélenchon wins under every one of the 24 dataset-strategy combinations. This provides strong support for the expectation that frontrunners are robust to the choice of missing data treatment, consistent with prior *Voter Autrement* findings (Baujard et al., 2014; Baujard, Igersheim & Lebon, 2024).

E2 – Lower-rank sensitivity (partially confirmed): Rank shifts do occur, but they are far smaller and rarer than anticipated. Only 4 of 48 candidate-dataset combinations show any sensitivity at all, and the affected candidates (Macron, Lassalle, and Arthaud) occupy middle ranks (6th to 8th) rather than the very bottom of the distribution. The candidates at the extremes of the ranking, both top and bottom, are completely stable. This suggests that sensitivity to imputation choice is concentrated specifically among candidates whose aggregate scores are closely clustered, rather than being a general property of lower ranks as such.

E3 – Lowest-rank bias (not supported by this dataset): No systematic disadvantage to less well-known candidates under lowest-rank imputation is observed in the rankings: candidates retain the same position under lowest-rank imputation as under every other strategy, including multiple imputation. This does not necessarily mean the theoretical concern underlying E3 is unfounded. Conflating unfamiliarity with disapproval remains a conceptually valid critique of lowest-rank imputation, but in this particular dataset, the practical consequences of that conflation were not large enough to alter any candidate's rank.

E4 – Multiple imputation as most defensible (not distinguishable in this dataset): While multiple imputation remains the most statistically principled approach on theoretical grounds, its rankings are observationally indistinguishable from those of the simpler strategies in this dataset. The theoretical advantage of MI, propagating uncertainty rather than substituting point estimates, did not manifest in different rank outcomes here, although it may do so in datasets with a higher proportion of missing values or closer electoral margins.

6.2 Interpreting the Robustness Findings

The main finding of this thesis is that the winner does not change depending on how missing values are handled. This is a reassuring result for the *Voter Autrement* study, since it suggests that their original choice of lowest-rank imputation did not distort the outcome. That said, it is important not to read too much into this finding. The most likely explanation is simply that Mélenchon's lead was large enough that no imputation method, however different its assumptions, could close the gap. In a closer election, the same six strategies could easily have pointed to different winners.

This conclusion is reinforced by the multiple imputation results in Section 5.3: because candidate rankings are computed independently within each of the five completed datasets, and pooled using Rubin's rules, the winner's stability is not an artefact of pooling but is confirmed directly at the level of each individual imputation, together with pooled confidence intervals that make the residual uncertainty from missing data explicit.

This has a practical implication for researchers working with similar datasets: the relevance of imputation choice depends heavily on how competitive the election is and how many values are missing. If the margin between candidates is narrow, the decision about how to fill in missing data is likely to have a real effect on the result. The takeaway from this thesis is therefore not that imputation does not matter, but that how much it matters depends on the data.

The small rank shifts observed among Macron, Lassalle, and Arthaud also deserve attention. These candidates are clustered closely in the middle of the ranking, and when scores are so close together, even minor differences in how missing values are filled in are enough to swap their positions. This fits with what E2 predicted: that the middle and lower tiers of a ranking are more sensitive to imputation choice than the top and bottom, where margins tend to be larger.

6.3 Limitations

Several limitations of the present analysis deserve explicit acknowledgment. First, the dataset is not representative of the French electorate. Participation in the *Voter Autrement* experiment was voluntary and self-selected, skewing towards politically engaged and highly educated individuals. The results therefore describe the preferences of a specific and non-random subset of voters and cannot be generalised to the wider population. This limits the normative conclusions that can be drawn about Range Voting as an electoral system, though it does not undermine the internal validity of the imputation comparison, which is the primary focus of the thesis.

Second, the analysis is restricted to a single election and a single national context. The 2022 French presidential election featured an unusually large margin of victory for the leading candidate under Range Voting, which is plausibly the main reason why no imputation strategy was able to alter the winner. Whether the same robustness would obtain in a more competitive field, or with a higher proportion of missing values, cannot be determined from this dataset alone.

Third, all six imputation strategies assume that the missing data mechanism is either MCAR or MAR. None of them is designed to handle MNAR, in which voters systematically avoid evaluating candidates they would have rated poorly. If MNAR is the true underlying mechanism, which cannot be ruled out in an electoral context, all six strategies may produce biased estimates, and the comparison between them would reflect the relative magnitude of that bias rather than its absence.

Finally, both the cluster-based and k-nearest-neighbour imputation strategies (Sections 4.6–4.7) default to k = 5. As Section 5.4 reports, robustness checks across k 3, 5, 7, 9 (cluster) and k 3, 5, 7, 9, 11 (kNN) confirm that Jean-Luc Mélenchon remains the winner and the full twelve-candidate ranking is unchanged in every combination tested; the choice of k therefore does not appear to be consequential within this range for this dataset. This does not rule out sensitivity at more extreme values of k, particularly for k approaching the size of the smallest demographic clusters (Section 4.6). Relatedly, because only four categorical demographic variables are available, the demographic clusters in Section 4.5 remain highly imbalanced, one large cluster dominates each dataset, which limits how much distinct subgroup information the strategy can exploit relative to a dataset with richer demographic covariates.

7 Conclusion

This thesis set out to answer a deceptively simple question: does imputation change the winner? Taking as its starting point the lowest-rank imputation applied in the original *Voter Autrement* 2022 experiment (Delemazure & Bouveret, 2024), six imputation strategies (lowest-rank, complete-case deletion, mean imputation, cluster-based imputation, k-nearest neighbour imputation, and multiple imputation) were applied to the four Range Voting datasets and compared against the stable Approval Voting and Borda Count baselines. In this case, the answer is no: across all 24 dataset-strategy combinations and both independent baselines, Jean-Luc Mélenchon wins regardless of how missing values are treated. Although the findings suggest that different imputation strategies do not affect the winning candidate in this dataset, this should not be interpreted as evidence that imputation choice is generally unimportant. The robustness observed here supports the validity of the lowest-rank imputation used in the original experiment, at least with respect to identifying the overall winner. However, this robustness is likely related to the size of Mélenchon's lead, which was large enough that the relatively small differences between imputation strategies did not alter the final outcome. The analysis also shows that this robustness is not absolute. A small number of mid-ranked candidates are affected by the choice of imputation method, suggesting that the impact of missing data treatment is greatest when candidates have very similar overall scores. In a closer election, or in a dataset with a higher proportion of missing values, the same six strategies might well have produced a different winner. The question addressed in this thesis remains relevant for any experimental electoral dataset, even when the results, as in this study, suggest that the choice of imputation strategy does not affect the outcome. Several paths for future research follow naturally from this work. The present analysis is limited to a single election and a single country; extending the comparison to earlier *Voter Autrement* waves or to electoral experiments in other national contexts would allow for an assessment of whether the robustness observed here is a general phenomenon or one specific to the 2022 French data and its particular margin of victory. Additionally, the relationship between non-response patterns and voter demographics, partially explored through cluster-based imputation, asks for deeper investigation, as understanding who leaves candidates unevaluated may be as informative as understanding how to fill those gaps. This conclusion is further supported by additional robustness analyses. Under multiple imputation, Jean-Luc Mélenchon remained the winner in every completed dataset generated for each Range Voting variant, indicating that winner stability is not merely a consequence of pooling but is preserved across repeated plausible completions of the missing data. Similarly, varying the tuning parameter k for both cluster-based and k-nearest neighbour imputation did not alter any candidate ranking, providing further evidence that the findings are robust to reasonable implementation choices. Ultimately, this thesis contributes to the understanding that rigorous electoral research demands transparency not only in the choice of voting rules, but in every methodological decision that precedes the final count. Missing data is not a peripheral concern; it sits at the intersection of statistical methodology and democratic representation, and it deserves to be treated accordingly.

8 References

- Barbaro, S. & Kurella, A.-S. (2025). Dichotomous Preferences: Concepts, Measurement, and Evidence. Working Papers 2506, Gutenberg School of Management and Economics, Johannes Gutenberg-Universität Mainz.

- Baujard, A., Igersheim, H. & Lebon, I. (2024). 2022 French Presidential Election: Different Voting Methods, Different Outcomes? Working Papers halshs-04841504, HAL.

- Baujard, A., Igersheim, H., Lebon, I., Gavrel, F. & Laslier, J.-F. (2014). Who's Favored by Evaluative Voting? An Experiment Conducted During the 2012 French Presidential Election. Electoral Studies, 34, 131–145.

- Baujard, A., Gavrel, F., Igersheim, H., Laslier, J.-F. & Lebon, I. (2018). How Voters Use Grade Scales in Evaluative Voting. European Journal of Political Economy, 55, 14–28.

- Delemazure, T. & Bouveret, S. (2024). Voter Autrement 2022 – The Online Experiment ("Un Autre Vote"). Dataset and companion article on Zenodo. https://doi.org/10.5281/zenodo.10

- Little, R. J. A. & Rubin, D. B. (2002). Statistical Analysis with Missing Data (2nd ed.).Wiley.

- van Buuren, S. (2018). Flexible Imputation of Missing Data (2nd ed.). Chapman & Hall/CRC. https://stefvanbuuren.name/fimd/

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